HOSPITAL DATA ANALYSIS USING SQL

Submitted by:

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Course:

Bachelor of Science in Information Technology (BSc IT)

University:

Mumbai University

Project Type:

30-Day SQL Micro Course

Tools & Technologies Used:

PostgreSQL (pgAdmin), Microsoft Excel

Academic Year:

2025–2026

Date of Submission:

23 March 2026

1. INTRODUCTION

In today's digital era, data plays a crucial role in decision-making across various industries, including healthcare. This project focuses on analyzing hospital data using Structured Query Language (SQL) in PostgreSQL.

The dataset contains important information such as hospital names, locations, departments, number of doctors, number of patients, admission and discharge dates, and medical expenses. By using SQL queries, meaningful insights can be extracted from this data.

The main objective of this project is to perform real-world data analysis using SQL concepts such as aggregation, grouping, sorting, and date calculations. This helps in understanding patient trends, hospital performance, and expense distribution.

This project provides practical exposure to handling structured data and improves problem-solving skills using database management systems.

2. DATASET DESCRIPTION

The dataset consists of the following columns:

1. Hospital Name:

   Represents the name of the hospital where the patient was treated.

2. Location:

   Indicates the city in which the hospital is located.

3. Department:

   Specifies the medical department such as Cardiology, Neurology, Orthopedics, etc.

4. Doctors Count:

   Shows the total number of doctors available in the respective hospital.

5. Patients Count:

   Represents the number of patients treated in a particular record.

6. Admission Date:

   The date on which the patient was admitted to the hospital.

7. Discharge Date:

   The date on which the patient was discharged from the hospital.

8. Medical Expenses:

   Indicates the total medical cost incurred for the treatment.

This dataset is used to perform various analytical operations such as aggregation, grouping, sorting, and date-based calculations to extract meaningful insights.

3. Data Import

The hospital dataset was initially prepared in Microsoft Excel and saved as a CSV (Comma-Separated Values) file. Before importing into PostgreSQL, the date format was standardized to 'YYYY-MM-DD' to ensure compatibility.

The dataset was then imported into PostgreSQL using pgAdmin's Import/Export feature. The table 'hospital_data' was created with appropriate data types, and the CSV file was successfully loaded into the database.

Below is a preview of the imported dataset:

Hospital Name	Location	Department	Doctors Count	Patients Count	Admission Date	Discharge Date	Medical Expenses
Wellness Clinic	Chennai	Orthopedics	47	182	13-12-2023	14-12-2023	31364.88
Fortis Care	Pune	ENT	15	51	29-12-2023	09-01-2024	47280.19
Wellness Clinic	Ahmedabad	Pediatrics	20	120	07-10-2023	21-10-2023	28574.72
Heritage Hospital	Hyderabad	Urology	8	172	29-04-2023	11-05-2023	7000.83
City Hospital	Kolkata	Gynecology	35	76	10-02-2023	12-02-2023	47210.46
Heritage Hospital	Hyderabad	Oncology	11	76	05-02-2023	17-02-2023	18612.34
Global Medicare	Ahmedabad	Oncology	22	99	01-08-2023	06-08-2023	47808.55
Apollo Health	Jaipur	General Medicine	37	173	01-10-2023	12-10-2023	12284.65
Heritage Hospital	Jaipur	ENT	9	198	12-06-2023	18-06-2023	14650.23
Green Valley Hospital	Pune	Pediatrics	30	107	07-12-2023	08-12-2023	30006.96
Healing Touch	Mumbai	Dermatology	39	70	13-09-2023	15-09-2023	13282.51
Fortis Care	Jaipur	Urology	22	38	12-03-2023	25-03-2023	13481.9
City Hospital	Lucknow	Cardiology	32	67	18-11-2023	24-11-2023	36748.42
Fortis Care	Jaipur	Orthopedics	27	60	18-09-2023	24-09-2023	37056.51
Sunrise Medical	Ahmedabad	Gynecology	48	37	06-08-2023	11-08-2023	15107.89
Healing Touch	Chennai	Cardiology	17	60	27-06-2023	30-06-2023	9348.55
Green Valley Hospital	Chennai	Pediatrics	40	96	22-03-2023	28-03-2023	48548.44
Healing Touch	Ahmedabad	Neurology	14	68	11-11-2023	22-11-2023	29942.75
Apollo Health	Delhi	Urology	41	111	05-11-2023	08-11-2023	43367.53
Fortis Care	Mumbai	Urology	46	154	14-02-2023	18-02-2023	41367.1
Metro Hospital	Bangalore	Dermatology	7	68	06-10-2023	07-10-2023	30136.26
Apollo Health	Hyderabad	Dermatology	28	191	08-04-2023	22-04-2023	17698.49
Fortis Care	Ahmedabad	Oncology	32	116	28-10-2023	30-10-2023	40799.02
Apollo Health	Hyderabad	Orthopedics	5	79	25-03-2023	29-03-2023	20952.51
City Hospital	Lucknow	Oncology	48	91	24-02-2023	04-03-2023	10882.4
Apollo Health	Pune	ENT	9	59	09-11-2023	16-11-2023	25177.79

Figure 1: The figure shows a preview of the hospital dataset in CSV format before importing into PostgreSQL.

3.1 Steps to Import Data into PostgreSQL

The hospital dataset was imported into PostgreSQL using pgAdmin by following these steps:

1. Opened pgAdmin and connected to the PostgreSQL server.

2. Created a new database named 'hospital_db'.

3. Created a table 'hospital_data' with appropriate column names and data types.

4. Right-clicked on the table and selected the "Import/Export Data" option.

5. Selected the CSV file containing hospital data.

6. Chose the format as CSV and enabled the "Header" option.

7. Set the delimiter as a comma (,) and executed the import process.

8. Verified the imported data using a SELECT query.

This process ensured that the dataset was successfully loaded into the PostgreSQL database for further analysis.

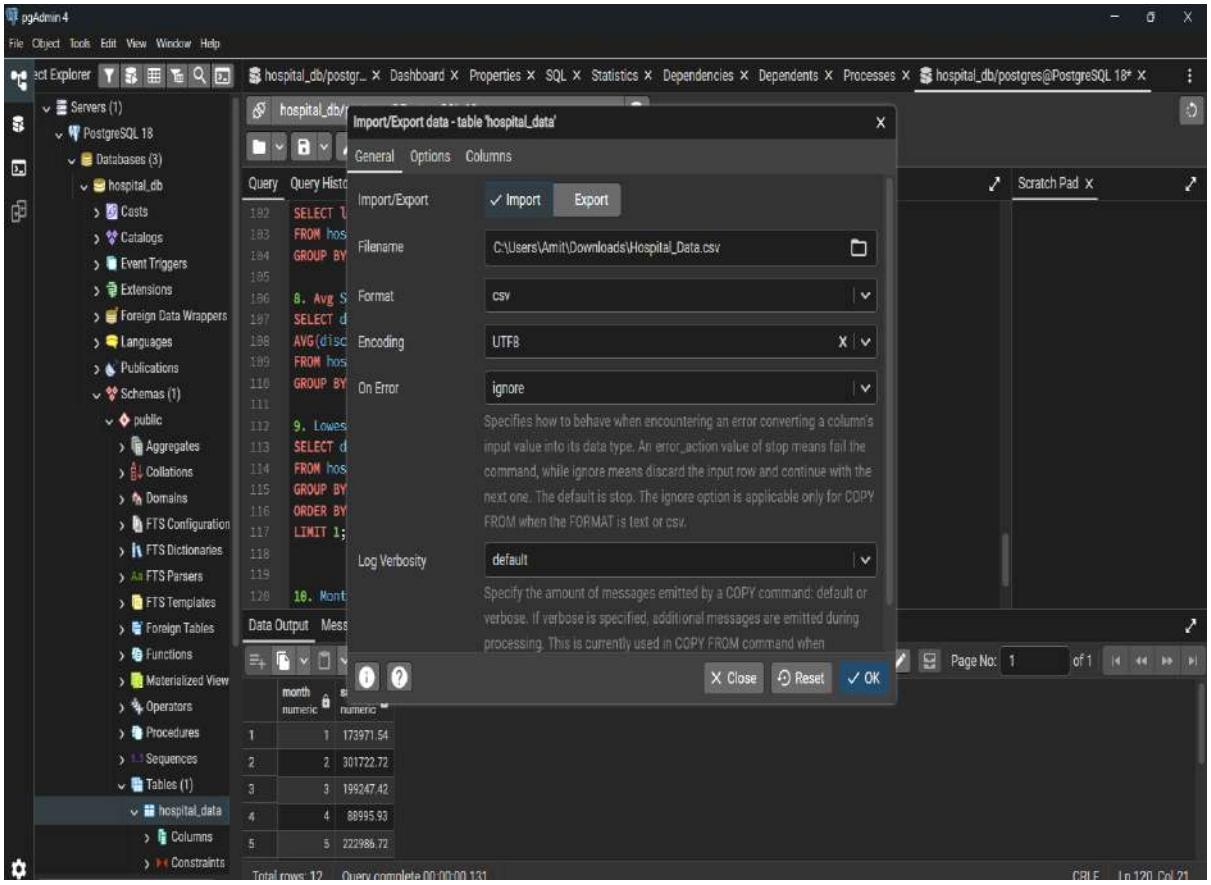


Figure 2: Importing CSV Data into PostgreSQL using pgAdmin

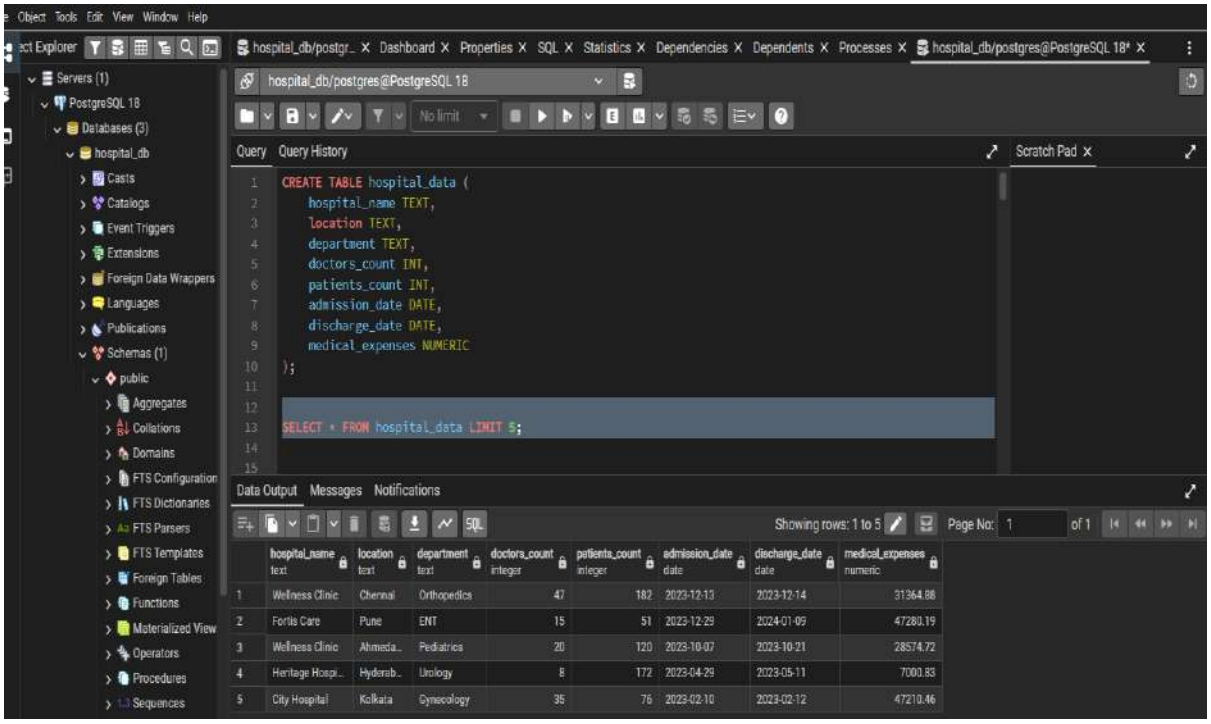


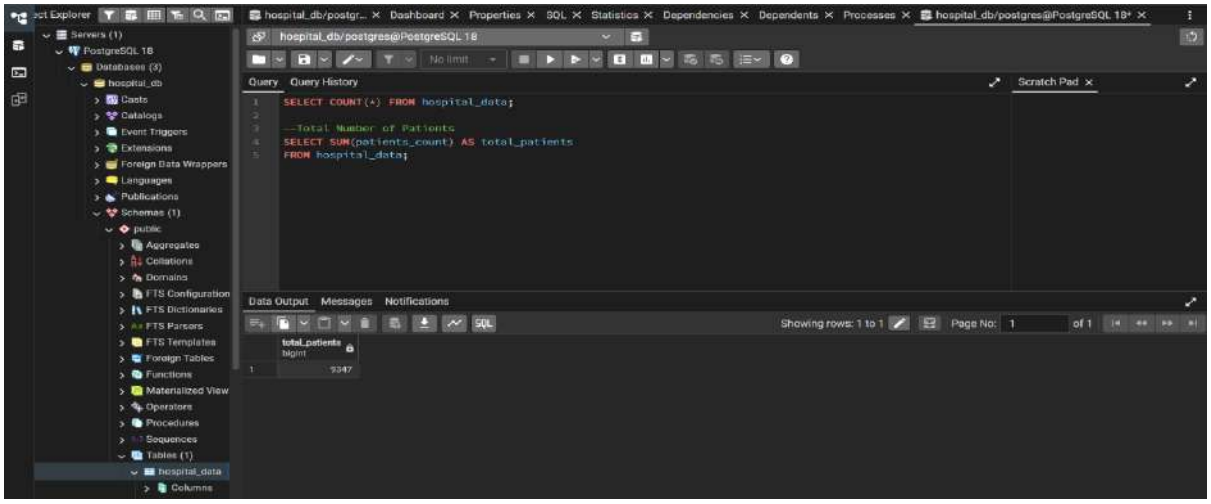
Figure 3:imported data successfully

SQL QUERIES AND OUTPUT

Query 1: Total Number of Patients

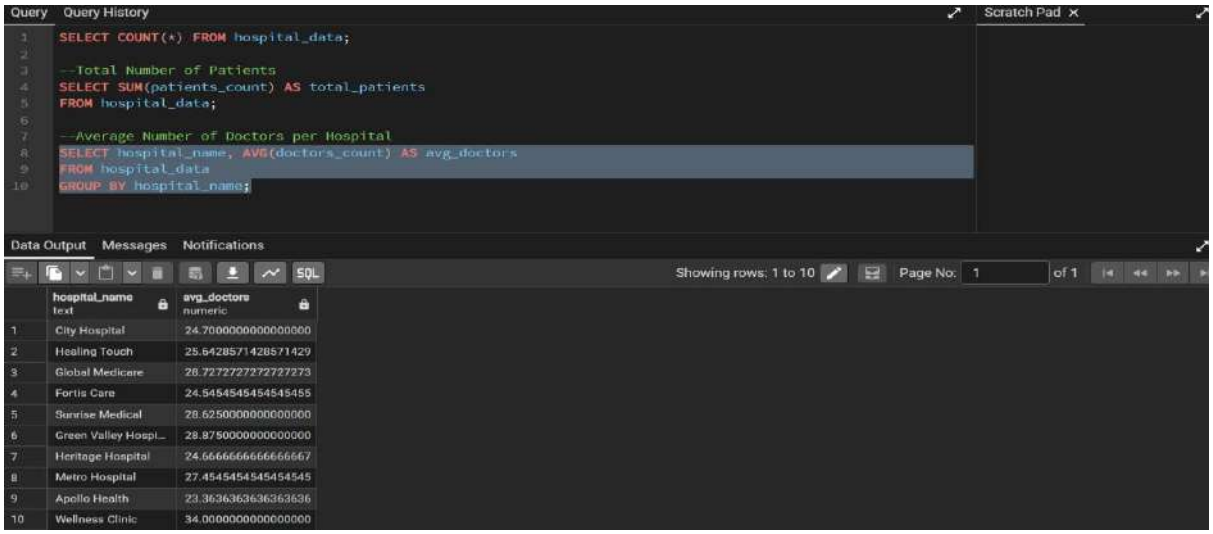
SELECT SUM(patients_count) AS total_patients

FROM hospital_data;



Query 2: Average Number of Doctors per Hospital

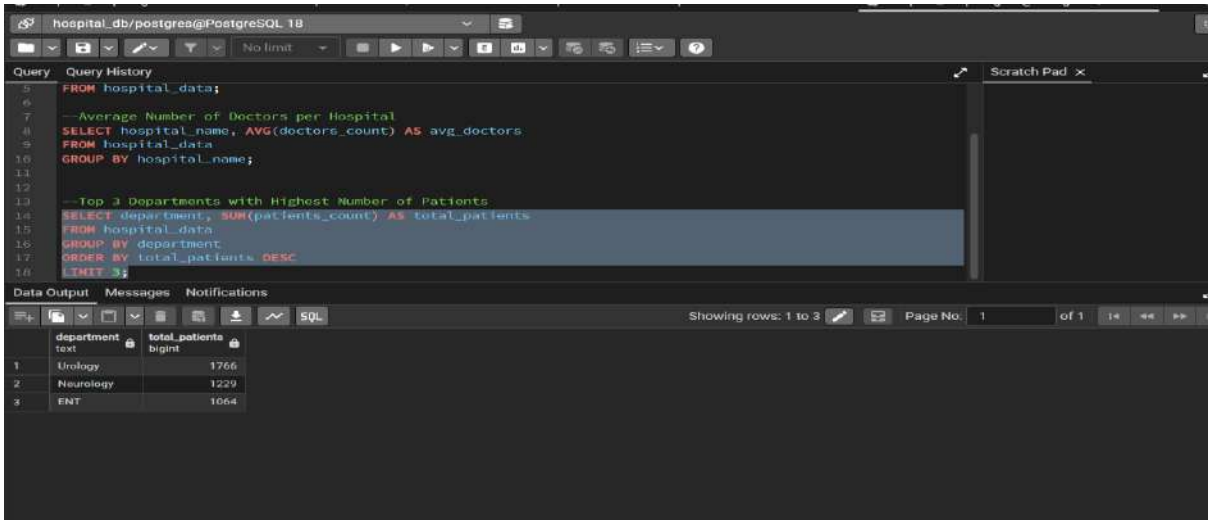
SELECT hospital_name, AVG(doctors_count) AS avg_doctors
FROM hospital_data
GROUP BY hospital_name;



	hospital_name text	avg_doctors numeric
1	City Hospital	24.7000000000000000
2	Healing Touch	25.6428571428571429
3	Global Medicare	28.7272727272727273
4	Fortis Care	24.5454545454545455
5	Sunrise Medical	28.6250000000000000
6	Green Valley Hospi...	28.8750000000000000
7	Heritage Hospital	24.6666666666666667
8	Metro Hospital	27.4545454545454545
9	Apollo Health	23.3636363636363636
10	Wellness Clinic	34.0000000000000000

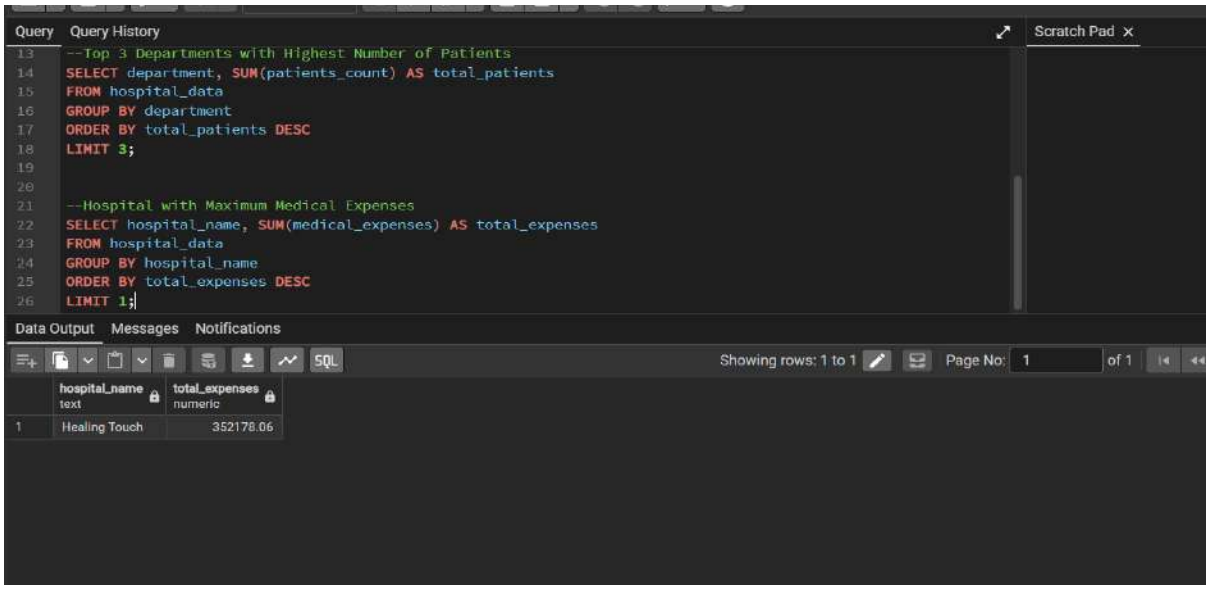
Query 3: Top 3 Departments with Highest Number of Patients

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients DESC
LIMIT 3;
```



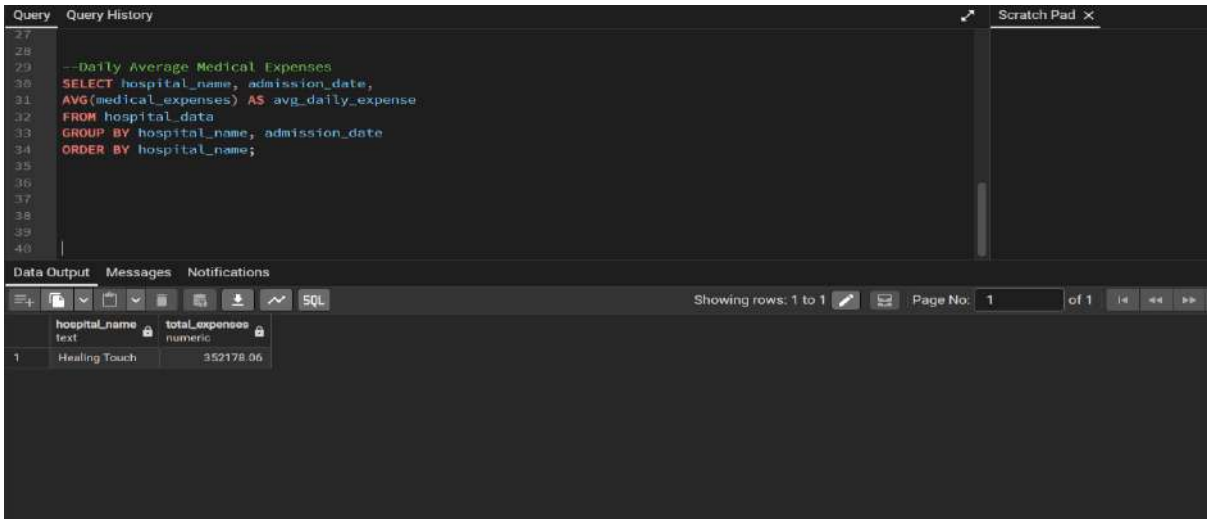
Query 4: Hospital with Maximum Medical Expenses

```
SELECT hospital_name, SUM(medical_expenses) AS total_expenses
FROM hospital_data
GROUP BY hospital_name
ORDER BY total_expenses DESC
LIMIT 1;
```



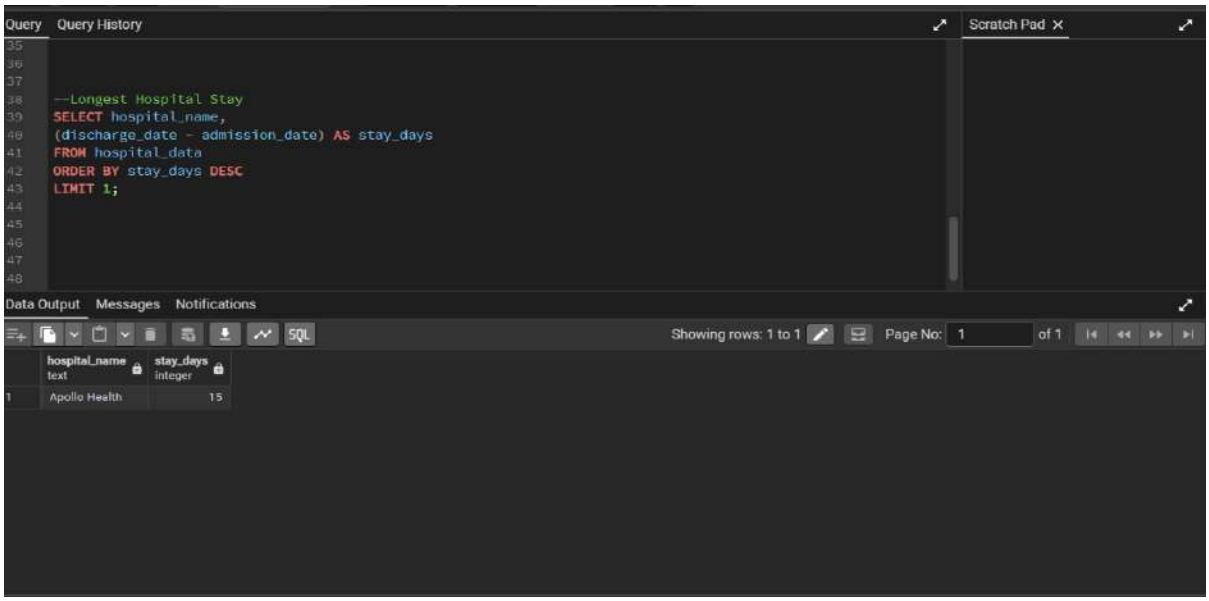
Query 5: Daily Average Medical Expenses

```
SELECT hospital_name, admission_date,
AVG(medical_expenses) AS avg_daily_expense
FROM hospital_data
GROUP BY hospital_name, admission_date;
```



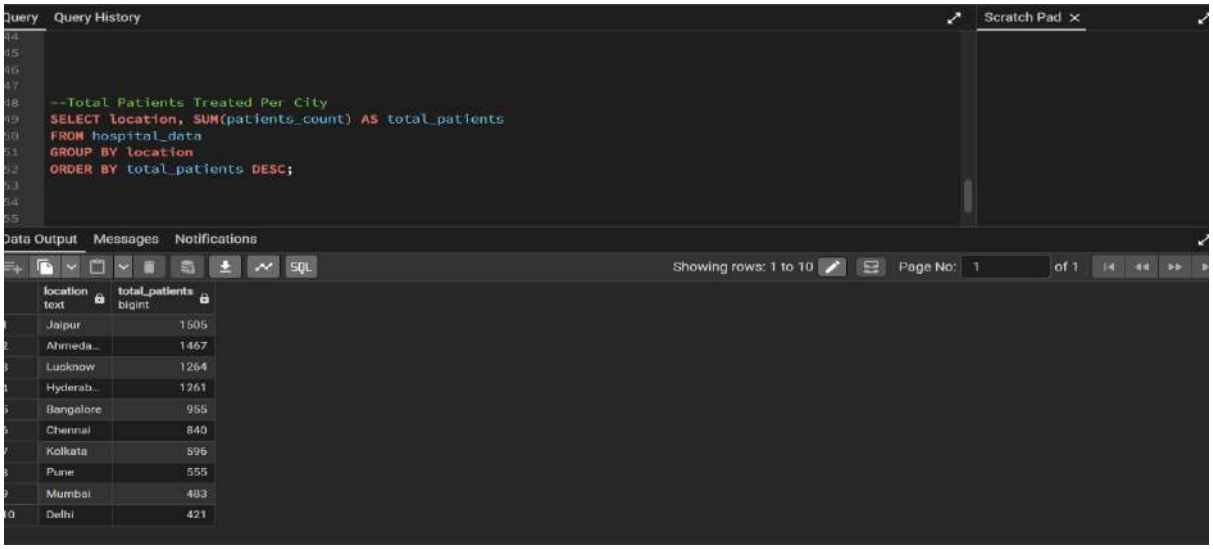
Query 6: Longest Hospital Stay

```
SELECT hospital_name,
(discharge_date - admission_date) AS stay_days
FROM hospital_data
ORDER BY stay_days DESC
LIMIT 1;
```



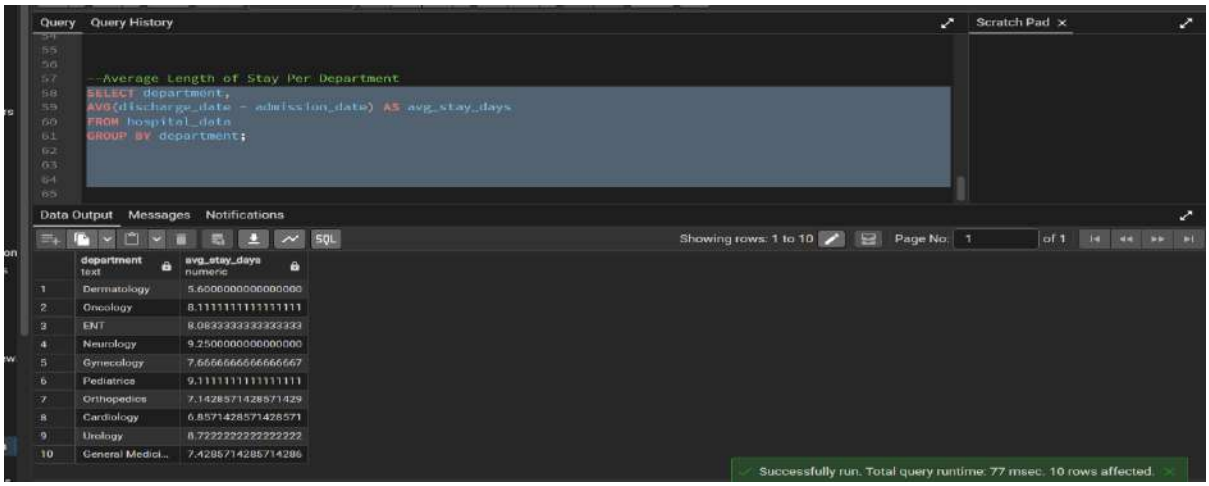
Query 7: Total Patients Treated Per City

SELECT location, SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY location;



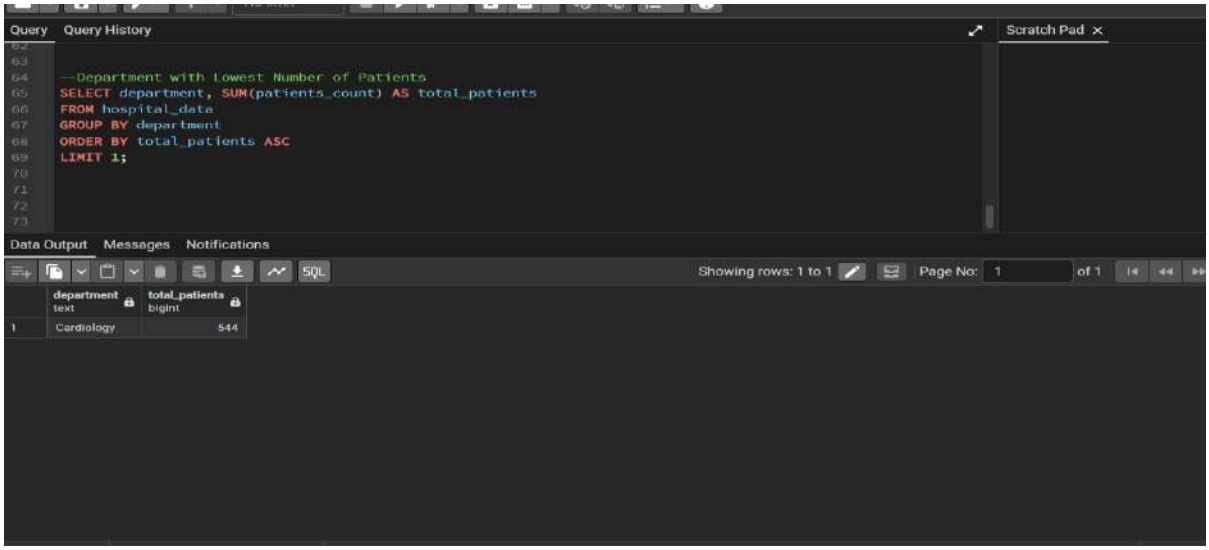
Query 8: Average Length of Stay Per Department

SELECT department,
AVG(discharge_date - admission_date) AS avg_stay_days
FROM hospital_data
GROUP BY department;



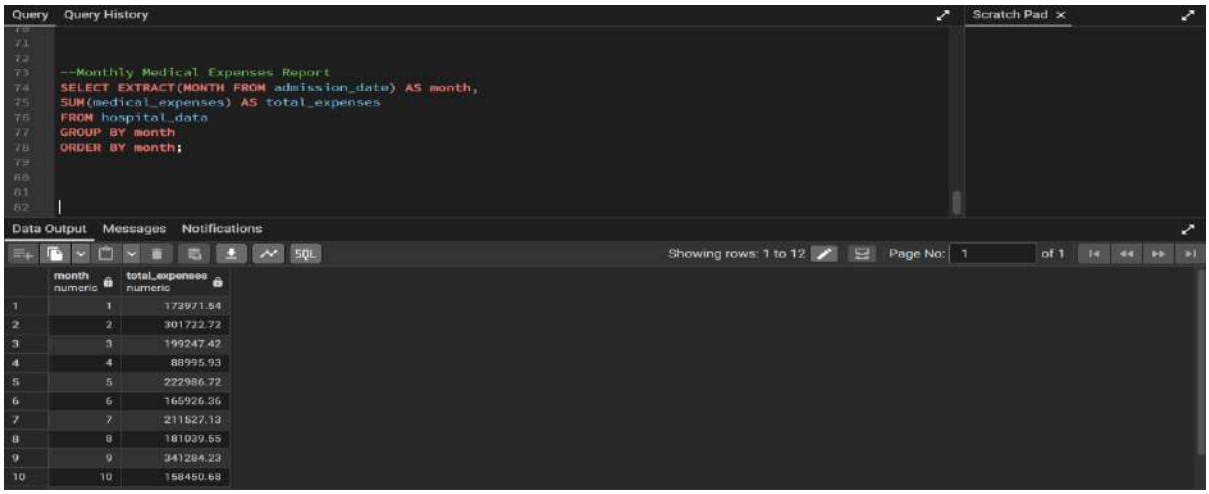
Query 9: Department with Lowest Number of Patients

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```



Query 10: Monthly Medical Expenses Report

```
SELECT EXTRACT(MONTH FROM admission_date) AS month,
SUM(medical_expenses) AS total_expenses
FROM hospital_data
GROUP BY month
ORDER BY month;
```



5. LEARNING OUTCOMES

Through this project, I gained the following skills and knowledge:

- **Learned how to import CSV data into PostgreSQL using pgAdmin**
- **Understood how to create and manage database tables**
- **Gained hands-on experience with SQL queries such as SELECT, GROUP BY, ORDER BY, and LIMIT**
- **Learned how to use aggregate functions like SUM and AVG**
- **Understood how to handle date operations and calculations**
- **Developed problem-solving skills using real-world datasets**
- **Improved debugging skills by resolving data type and import errors**

This project helped in building a strong foundation in SQL and data analysis.